

# Sonal Agarwal

Davis, California — [sagarwa@ucdavis.edu](mailto:sagarwa@ucdavis.edu) — (510) 674-8357 — U.S Citizen

## Education

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**University of California, Davis**, Davis, California  
Environmental Science and Management, B.S  
*GIS and Environmental Data Science Track*

Enrolled: Sept. 2022 – Expected: June 2026  
GPA: 3.78

## Experience

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### **Geospatial Data Student Assistant – UC Davis Datalab**

*UC Davis, CA – January 2025 – June 2025*

- Digitized American Viticultural Areas (AVAs) in QGIS to help the development of viticultural research and industry.
- Analyzed topographic maps, followed detailed instructions, and filled attribute tables with metadata to create complex polygons.
- Assisted with project organization on GitHub, including monitoring for digitization issues and updating QGIS instructions relevant to the project.
- Developed R code to facilitate the cleaning, management, and categorization of AVA files in GeoJSON format.

### **Agrivoltaics Research Intern – Dr. Majdi Abou Najm Research Lab**

*UC Davis, CA – April 2024 – October 2024*

- Conducted field measurements of tomato and pepper growth under varying solar conditions. Handled instruments that measured infrared radiation, chlorophyll content, photosynthetically active radiation, and soil moisture.
- Managed all excel data related to the project and created R data visuals, analyses, and presentations highlighting plant growth progress, photosystem efficiency, stomatal conductance, electromagnetic radiation, etc. R visuals include box-and-whisker plots, scatterplots, and bar graphs.
- Conducted online research using the IRA to find government incentives and benefits in underrepresented farming communities for the implementation of agrivoltaics. Examined GIS maps of energy communities in California to understand possible methods of building a GIS map.

### **Environmental Educator – Sprout Up**

*Davis, CA – October 2023 - December 2023*

- Volunteered as an environmental educator at Patwin Elementary School in Davis, CA. Integrated the importance of sustainability and environmental science in classrooms to about 25 first-grade students.
- Worked closely with diverse groups of 5-6 students in each classroom explaining complex concepts in simplified ways and monitored the growth of the student group.
- Showed leadership by communicating with peer volunteers, lead volunteers, and teachers; easily adapted to lesson plans and schedules.

## Skills

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### **Relevant Coursework**

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|----------------------------------------------|----------------------------------------|
| • Introduction to GIS                        | • Nonparametric Statistics             |
| • Concepts/Methods in GIS                    | • Environmental Monitoring             |
| • Environmental Policy                       | • Unmanned Aerial Systems              |
| • Remote Sensing and Aerial Photography      | • Urban and Regional Planning          |
| • Applied Statistics and Regression Analysis | • Introduction to Programming (Python) |

### **Relevant Skills**

- Experience in using Excel for data organization and graphing, RStudio for statistical analysis and geospatial data analysis using terra and sf packages, and Python.
- Experience in ArcGIS Pro, QGIS, and Google Earth Pro for mapping and analysis.
- Experience in ENVI and Pix4D for aerial image processing, visualization, and interpolation.

**GitHub Link:** <https://github.com/s0nalagarwal>